

SUKHWINDER SINGH

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EDUCATION

Purdue University - West Lafayette
Bachelor's, Cybersecurity

August 2023 - May 2026

CERTIFICATIONS

CompTIA Security+ | CompTIA Network+ (In Progress) | Tata Cybersecurity Analyst Job Simulation (IAM)

PROFESSIONAL EXPERIENCE

Purdue University - West Lafayette
Network & Systems Support Intern

July 2025 - July 2026
Indianapolis, IN, USA

- Monitored LAN/WAN health across campus devices using centralized dashboards, tracking bandwidth, latency, and uptime to catch issues before they reached end users.
- Configured firewall rules and network access controls, and reviewed traffic logs to spot misconfigurations and suspicious activity.
- Triaged alerts and support tickets, troubleshoot TCP/IP, DNS, DHCP, and VPN issues, and escalated security incidents with documented root cause.
- Applied access control and encryption measures across network endpoints to protect data in transit, in line with university security and compliance standards.
- Used AI tools with Python and PowerShell scripting to automate routine checks, log summaries, and incident documentation.

2020 Companies (Supporting Meta)
IT Systems Intern

January 2024 - July 2025
Carmel, IN, USA

- Monitored system activity and logs to flag performance issues and potential security threats, escalating findings to the IT security team.
- Supported vulnerability management by diagnosing issues, deploying patches and critical updates, and tracking open risks through remediation.
- Troubleshoot enterprise hardware and software issues through the ticketing queue, documenting fixes and following security protocols.

PROJECTS & OUTSIDE EXPERIENCE

Network Monitoring with Zabbix & PRTG (Purdue Living Lab)

- Installed Zabbix 7.0 on Linux and configured SNMP on a production server, taking a live server room from no monitoring to 235 tracked metrics with email alerting.
- Solved a VM networking blocker by bridging a second NIC onto the server subnet, then opened only UDP/161 in iptables to follow least privilege.
- Built PRTG Ping, DNS, and HTTP sensors to baseline latency, name resolution, and service availability for faster root cause analysis.

pfSense Firewall & Network Segmentation

- Configured pfSense across WAN, LAN, and DMZ zones, allowing web traffic into the DMZ while blocking DMZ-to-LAN movement.
- Set up DHCP reservations and BIND DNS zones, and disabled unused services to reduce the attack surface.
- Validated rules through testing, packet inspection, and firewall log review to confirm isolation.

Endpoint Security Monitoring & Log Analysis

- Monitored endpoint and network activity with Snort and Suricata IDS, investigating alerts to separate real threats from noise.
- Simulated attacks on Ubuntu and mapped activity to MITRE ATT&CK to tune detection rules and incident response steps.
- Built a Python tool to triage a 44K-event firewall log, flagging denied connections and source countries for faster SOC review.

SKILLS

Skills: Active Directory, Windows Server, Windows 10/11, Linux (Ubuntu, Kali, CentOS), Microsoft 365, TCP/IP, DNS, DHCP, VPN, LAN/WAN, VLANs, Firewalls (pfSense, iptables), SIEM (Splunk), IDS/IPS (Snort, Suricata), Wireshark, Nmap, OpenVAS, Zabbix, PRTG, SNMP, Alert Triage, Incident Response, Vulnerability & Patch Management, Log Analysis, MITRE ATT&CK, Access Control (IAM), Ticketing & Documentation, Root Cause Analysis, Python, PowerShell, Bash, SQL, AI-Assisted Automation
Languages: Hindi, Punjabi, Spanish, Urdu